HAWK Fakultät Ingenieurwissenschaften und Gesundheit Göttingen

POLITECNICO MILANO 1863

Worked Examples

- including solutions -

ADVANCED TOPICS IN ENGINEERING PHYSICS

'Plasma and its applications'

Worked example 1

Task

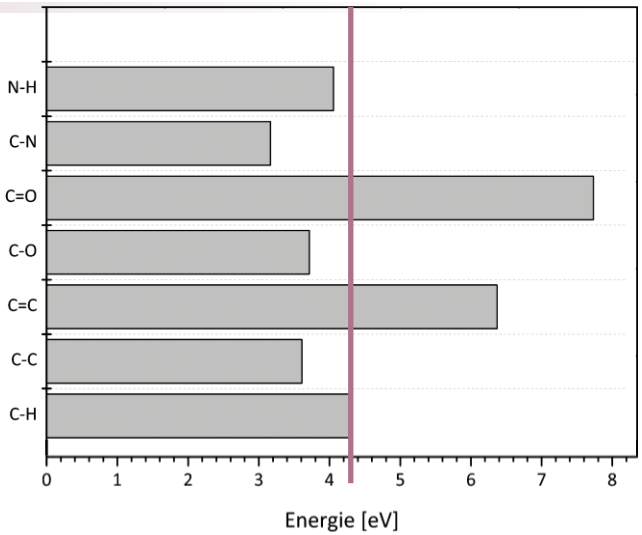
- For surface cleaning purposes, the high voltage electrode of a DC-DBD plasma source, which is operated at atmospheric pressure, is located above a work piece made of pure aluminium. This work piece acts as counter electrode where the plasma discharge thickness is 2 mm. The aim is to remove a monolayer of grease (consisting of -C-H- single bonds) from the work piece via electron impact-induced dissociation.

  a) Estimate the required voltage which has to be applied to the electrode for obtaining such cleaning.

  b) Which plasma process gas should be used for the above-described application?

Worked example 1

$E_{-C-H-} \approx 4.25eV$



Solution a

The bond strength of -C-H- is about 4.25 eV:

The kinetic energy of the electron should thus be at least the same value.

$V = ?$ $\qquad V = \vec{E} \cdot d$

$\vec{E} = ?$ $\qquad E_{kin} = \vec{E} \cdot e \cdot \lambda$

$$E_{kin} \approx 4.25eV = 6.81 \cdot 10^{-18} J$$

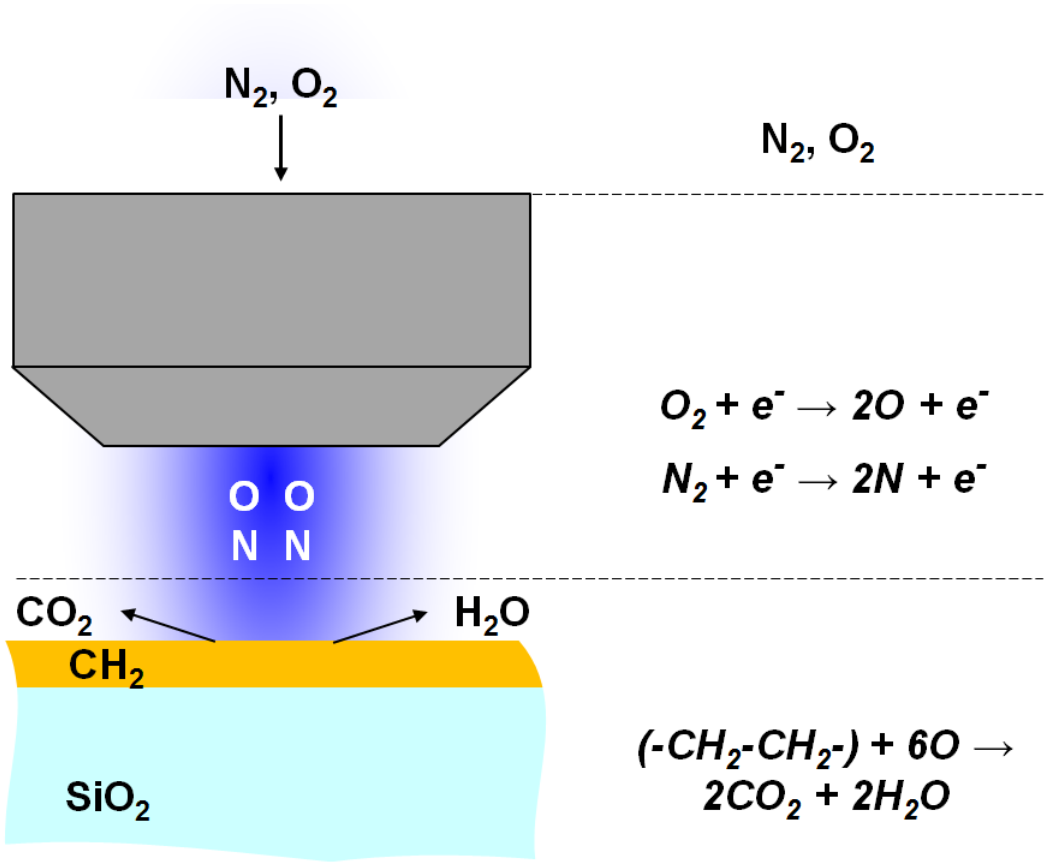
$$\rightarrow \vec{E} = \frac{E_{kin}}{e \cdot \lambda} = \frac{6.81 \cdot 10^{-18} VAs}{1.6 \cdot 10^{-19}\, As \cdot 68 \cdot 10^{-9}\, m} = 626 MV/m$$

$$\rightarrow V = \vec{E} \cdot d = 626 MV/m \cdot 0.002m = 1.252 kV$$

Worked example 1

Solution b

To support electron impact-induced breaking (= plasma-physical mechanism) of the -C-H- bond by plasma-chemical interactions, air should be chosen/favoured as process gas.



$$O_2 + e^- \rightarrow 2O + e^-$$

$$N_2 + e^- \rightarrow 2N + e^-$$

$$(\text{-}CH_2\text{-}CH_2\text{-}) + 6O \rightarrow 2CO_2 + 2H_2O$$

Worked example 1

Alternative consideration

- What if the cleaning process was carried out in a low pressure chamber under rough vacuum of 1 hPa…?

Pressure range	**Pressure *p* in hPa**	**Particle density *n* in n/cm³**	**Mean free length of path* λ**
Atmospheric	1013	$12{,}7 \times 10^{19}$	68 nm
Rough vacuum	300 – 1	$10^{19} – 10^{16}$	0,1 – 100 µm
Fine vacuum	$1 – 10^{-3}$	$10^{16} – 10^{13}$	0,1 – 100 mm
Full vacuum	$10^{-3} – 10^{-7}$	$10^{13} – 10^{9}$	10 cm – 1 km

$$\rightarrow \vec{E} = \frac{E_{kin}}{e \cdot \lambda} = \frac{6.81 \cdot 10^{-18} VAs}{1.6 \cdot 10^{-19}\, As \cdot 1 \cdot \boxed{100^{-6}\, m}} = 626 kV/m$$

$$\rightarrow V = \vec{E} \cdot d = 626 kV/m \cdot 0.002 m = 1.3 kV$$

Required voltage is reduced by same factor as mean free length of path

Why?

Particle density and thus number of decelerating collision events is reduced.

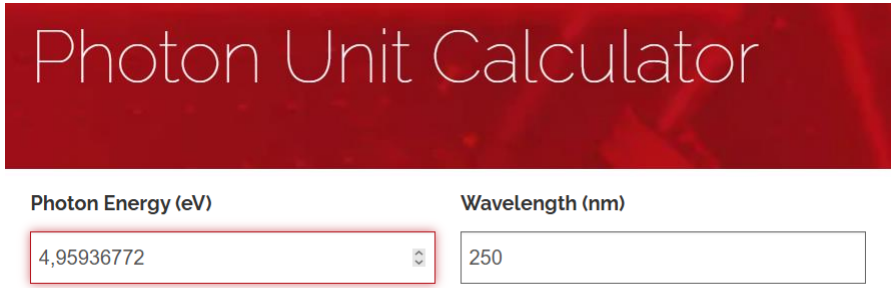
Worked example 2

Task

- In a high-power plasma generator, which can be described as a plane-parallel capacitor, the voltage applied to the electrodes made of copper is 10 kV. The distance between the electrodes is 1 mm. One copper electrode of this setup is irradiated by incident cosmic irradiation at a wavelength of 250 nm. The work function of the copper electrode material is 5.0 eV.

- Is an electron released from the electrode material by the incident irradiation in the given case?

Worked example 2



Solution (1 of 2)

At a first glance, one could say 'no', since $E_{photon} < W$

BUT:
The work function merely refers to uncharged surfaces!

In the present case, the electrode surface is subject to an electric field due to the applied voltage:

$$V = \vec{E} \cdot d \rightarrow \vec{E} = \frac{V}{d} = \frac{10kV}{1mm} = 10MV/m$$

This comes along with a decrease in work function ΔW due to the Schottky-effect according to

$$\Delta W = \sqrt{\frac{e^3 \cdot \vec{E}}{4 \cdot \pi \cdot \varepsilon_0}}$$

Worked example 2

Solution (2 of 2)

$$\Delta W = \sqrt{\frac{e^3 \cdot \vec{E}}{4 \cdot \pi \cdot \varepsilon_0}}$$ for $\vec{E} = 10MV/m$ amounts to

$$\Delta W \approx 0.12eV$$

The effective work function of the charged electrode surface is thus

$$W_{eff} = W - \Delta W \approx 5.0eV - 0.12eV = 4.88eV$$

Hence, the photon energy of about 4.96 eV provided by the incident cosmic irradiation is sufficient for releasing elctrons from the charged electrode material.

Worked example 3

Task

- A high-pressure LTE plasma is ignited between two plane and parallel electrodes where the discharge gap thickness is 1 mm. The process gas in the discharge gap is argon, the total particle density is $10^{30}/m^3$.

  a) Determine the required temperature of this plasma for obtaining ionisation of the process gas via electron impact-induced ionisation neglecting any acceleration of electrons by an electric field.

  b) Determine the required voltage that should be applied to the electrodes for obtaining ionisation of the process gas via electron impact-induced ionisation for the plasma at a temperature of 10,000K, however neglecting any temperature-driven acceleration of electrons.

Worked example 3

Ionization energy determinations

IE (eV)	Method	Reference	Comment
15.763	PIPECO	Weitzel, Mahnert, et al., 1994	*LL*
15.75962	EVAL	Lide, 1992	*LL*
15.82	EI	Wetzel, Baiocchi, et al., 1987	*LBLHLM*
15.760	S	Kelly, 1987	*LBLHLM*
15.759	PE	Kimura, Katsumata, et al., 1981	*LLK*
15.88	EI	Clare and Sowerby, 1981	*LLK*
15.7	EI	Freiser, 1980	*LLK*
15.75962 ± 0.00001	S	Minnhagen, 1973	*LLK*
15.753 ± 0.002	TE	Spohr, Guyon, et al., 1971	*LLK*
15.75962 ± 0.00001	S	Yoshino, 1970	*RDSH*
15.759	S	Yoshino, 1969	*RDSH*
15.713 ± 0.003	CI	Hotop and Niehaus, 1969	*RDSH*
15.757 ± 0.005	PE	Collin and Natalis, 1968	*RDSH*
15.74 ± 0.05	EI	Gallegos and Klaver, 1967	*RDSH*
15.78 ± 0.03	EI	Winters, Collins, et al., 1966	*RDSH*
15.79	PE	Al-Joboury and Turner, 1963	*RDSH*

Solution a (1/2)

For electron impact-induced ionisation neglecting any temperature-driven acceleration (= thermal drift), the kinetic electron energy,

$$E_{kin} = \frac{1}{2} \cdot m \cdot v^2,$$

should be equal to the ionisation energy of the gas. In the case of argon with E_i = 15.88 eV = $2.54 \cdot 10^{-18}$ J (worst case in list in NIST database, see table above taken from: https://webbook.nist.gov/cgi/inchi?ID=C7440371&Mask=20), the electron velocity should thus be

$$v = \sqrt{\frac{2 \cdot E_{kin}}{m}} = \sqrt{\frac{2 \cdot 2.54 \cdot 10^{-18} J}{9.11 \cdot 10^{-31} kg}} = 2{,}361{,}417 \frac{m}{s}.$$

Worked example 3

Solution a (2/2)

In a high-pressure LTE plasma, local Maxwell–Boltzmann distribution occurs where the most probable velocity of an electron is given by

$$v_{max} = \sqrt{\frac{2 \cdot k_B \cdot T}{m}}.$$

The required LTE plasma temperature is thus

$$T = \frac{v_{max}^2 \cdot m}{2 \cdot k_B} = \frac{(2{,}361{,}417 m/s)^2 \cdot 9.11 \cdot 10^{-31} kg}{2 \cdot 1.38 \cdot 10^{-23} \frac{m^2 \cdot kg}{s^2 \cdot K}} = 184{,}057 K.$$

Worked example 3

Solution b

For solving this task, the mean free length of path of the high-pressure argon plasma needs to be determined initially. It is given by

$$\lambda = \frac{1}{n \cdot \sigma} = \frac{1}{n \cdot \pi \cdot (r_1 + r_2)^2}.$$

The theoretical electron radius is 2.8 x 10^{-15} m (remember: estimation, since sizes of quantum objects cannot be determined definitely) and the mean radius of an argon atom is 71 pm, leading to

$$\lambda = \frac{1}{10^{30}/m^3 \cdot \pi \cdot (2.8^{-15}m + 71^{-12}m)^2} = 63pm.$$

The required energy for ionisation is $2.54 \cdot 10^{-18}$ J, leading to a required electric field strength within the discharge gap of

$$\vec{E} = \frac{E_{kin}}{e \cdot \lambda} = \frac{2.54 \cdot 10^{-18} VAs}{1.6 \cdot 10^{-19}\, As \cdot 63 \cdot 10^{-12}\, m} = 251984 MV/m.$$

The voltage to be applied is thus

$$V = \vec{E} \cdot d \approx 252MV.$$

…which is extraordinary high, demonstrating that ionisation of a process gas in such a plasma is not only achieved by pure kinetic, but also thermal impact.

Worked example 4

Task

- A plasma source consists of two plane-parallel electrodes with an overall discharge gap of 18 mm and is operated at atmospheric pressure. The applied alternating voltage amounts to 11 kV. Via incident cosmic irradiation, an electron is generated in the gas-filled discharge gap at a distance of 100 µm from ~~the high voltage electrode..~~ one of the electrodes.

  a) Which high-voltage pulse frequency should be applied to ‘trap’ the electron between the electrodes…so that it does not impact on an electrode surface over a longer operation time of some minutes?

  b) Determine both the velocity and kinetic energy of the electron during oscillation as resulting from a).

Worked example 4

Solution a (1/2)

- In order to avoid that the electron hits the opposite electrode, we define the same distance of 100 µm from the second electrode. The oscillation range of the electron between the electrodes is thus 18 mm – 200 µm = 17.8 mm. Its kinetic energy is given by $E_{kin} = \frac{1}{2} \cdot m \cdot v^2$ (1) and $E_{kin} = \vec{E} \cdot e \cdot \lambda$ (2), respectively. Moreover, its velocity follows from $v = \frac{s}{t}$ (3).
- The time of flight (*TOF*) of the electron can then be determined by inserting (2) in (1) and inserting the resulting expression in (3) and solving (3) for *t* = *TOF*:

$$v = \frac{s}{t} \rightarrow t = TOF = \frac{s}{v} = \frac{s}{\sqrt{\frac{2 \cdot E_{kin}}{m}}} = \frac{s}{\sqrt{\frac{2 \cdot \vec{E} \cdot e \cdot \lambda}{m}}} = \frac{s}{\sqrt{\frac{2 \cdot \frac{V}{d} \cdot e \cdot \lambda}{m}}}$$

- Note: $\vec{E} = \frac{V}{d}$ and λ = 68 nm (atmospheric pressure)

Worked example 4

Solution a (2/2)

- The time of flight is thus

$$TOF = \frac{s}{\sqrt{\frac{2 \cdot \frac{V}{d} \cdot e \cdot \lambda}{m}}} = \frac{0.0178m}{\sqrt{\frac{2 \cdot \frac{11000V}{0.018m} \cdot 1.6 \cdot 10^{-19}\,As \cdot 68 \cdot 10^{-9}m}{9.11 \cdot 10^{-31}kg}}}$$

$$TOF = 1.47 \cdot 10^{-7}s = 147ns$$

Since this time corresponds to half the oscillation period time *T* (there and back again!), the required frequency of the high-voltage pulses is finally

$$f = \frac{1}{T} = \frac{1}{2 \cdot TOF} \approx 3.4MHz$$

Worked example 4

Solution b

- The velocity of the electron is

$$v = \frac{s}{t} = \frac{0.0178m}{147 \cdot 10^{-9}s} = 121088\frac{m}{s}$$

- Its kinetic energy is

$$E_{kin} = \frac{V}{d} \cdot e \cdot \lambda = \frac{11000V}{0.018m} \cdot 1.6 \cdot 10^{-19}\,As \cdot 68 \cdot 10^{-9}m = 6.65 \cdot 10^{-21}J$$

- Control:

$$E_{kin} = \frac{1}{2} \cdot m \cdot v^2 = \frac{1}{2} \cdot 9.11 \cdot 10^{-31}kg \cdot \left(121088\frac{m}{s}\right)^2 = 6.6510^{-21}J$$

…corresponding to about 0.04 eV.

Worked example 4

Alternative consideration

- How does the kinetic energy of the electron change if pressure is reduced?

Pressure range	**Pressure *p* in hPa**	**Particle density *n* in n/cm³**	**Mean free length of path* λ**
Atmospheric	1013	$12{,}7 \times 10^{19}$	68 nm
Rough vacuum	300 – 1	$10^{19} – 10^{16}$	0,1 – 100 µm
Fine vacuum	$1 – 10^{-3}$	$10^{16} – 10^{13}$	0,1 – 100 mm
Full vacuum	$10^{-3} – 10^{-7}$	$10^{13} – 10^{9}$	10 cm – 1 km

For a constant electric field strength, the kinetic energy increases linearly with decreasing pressure and increasing mean free length of path, respectively, according to $E_{kin} = \vec{E} \cdot e \cdot \lambda$.